

# SOFTWARE ENGINEERING

## Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

### Digital Campus Library Reservation Gateway

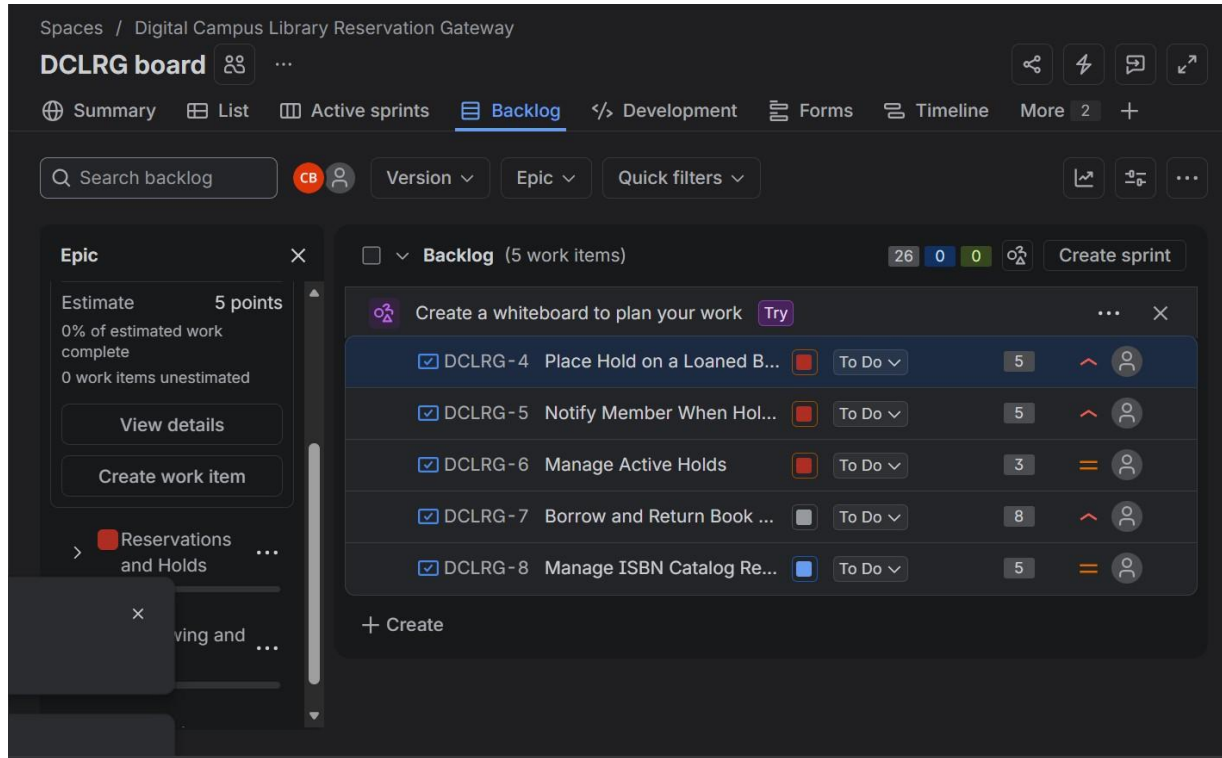
Jira Scrum Backlog and Sprint Report

|                          |                                                  |
|--------------------------|--------------------------------------------------|
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| <b>SRN</b>               | PES1UG24CS123                                    |
| <b>Section</b>           | B                                                |
| <b>Course</b>            | UE24CS341A : Software Engineering                |
| <b>Domain</b>            | Campus & Academic Operations                     |
| <b>Problem Statement</b> | #08 - Digital Campus Library Reservation Gateway |
| <b>Jira link</b>         | <a href="#">DCLRG board - Backlog - Jira</a>     |

This report documents the Jira backlog, story-point estimation, sprint execution, completion evidence, and Jira analytics for the Digital Campus Library Reservation Gateway.

## 1. Epics and Backlog Structure

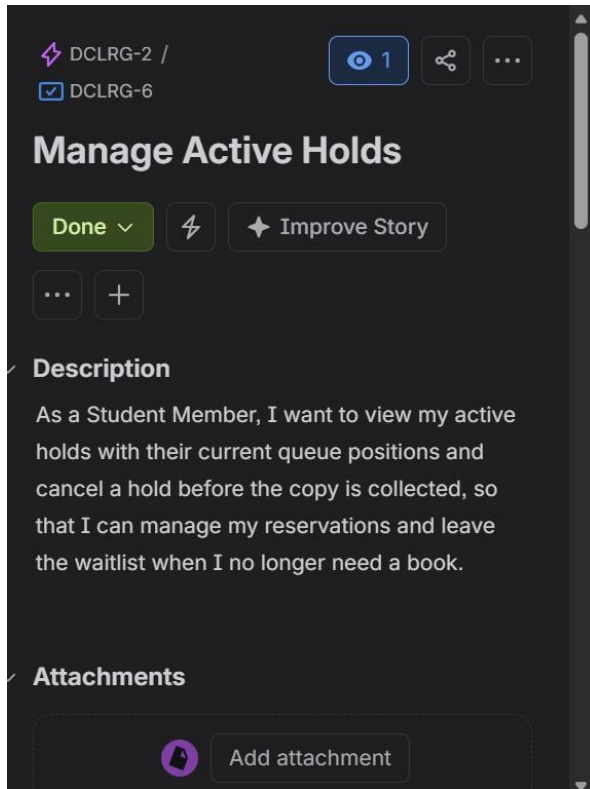
The Jira project was organized around three functional themes derived from the library reservation scenario.



| Epic                   | User stories represented in the backlog                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Catalog Management     | DCLRG-8 - Manage ISBN Catalog Records                                                                                         |
| Reservations and Holds | DCLRG-4 - Place Hold on a Loaned Book<br>DCLRG-5 - Notify Member When Hold Becomes Available<br>DCLRG-6 - Manage Active Holds |
| Borrowing and Fines    | DCLRG-7 - Borrow and Return Book with Fine Tracking                                                                           |

## 2. User Story Details

The selected completed stories retain the user-story format and show their final Done status in Jira.



✦ DCLRG-2 /  
✓ DCLRG-6

**Manage Active Holds**

Done ▾ ⚡ ✦ Improve Story

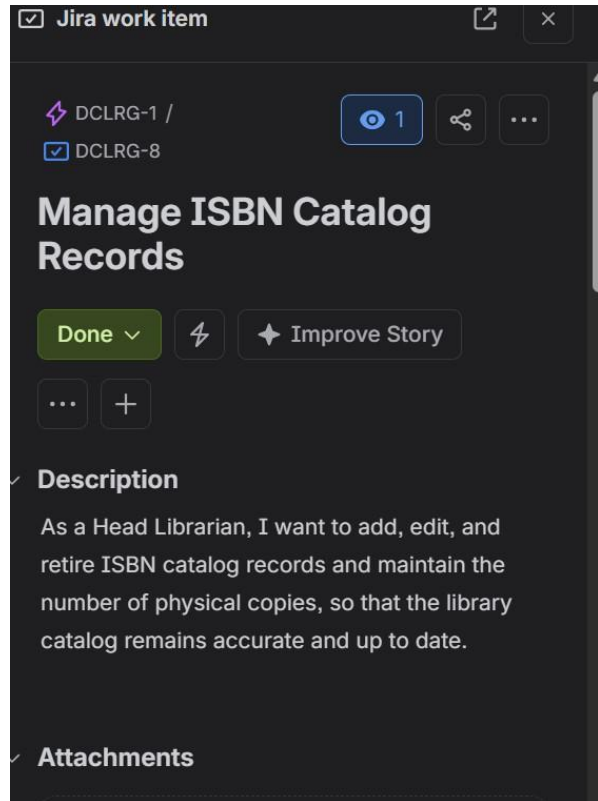
⋮ +

**Description**

As a Student Member, I want to view my active holds with their current queue positions and cancel a hold before the copy is collected, so that I can manage my reservations and leave the waitlist when I no longer need a book.

**Attachments**

 Add attachment



✓ Jira work item

✦ DCLRG-1 /  
✓ DCLRG-8

**Manage ISBN Catalog Records**

Done ▾ ⚡ ✦ Improve Story

⋮ +

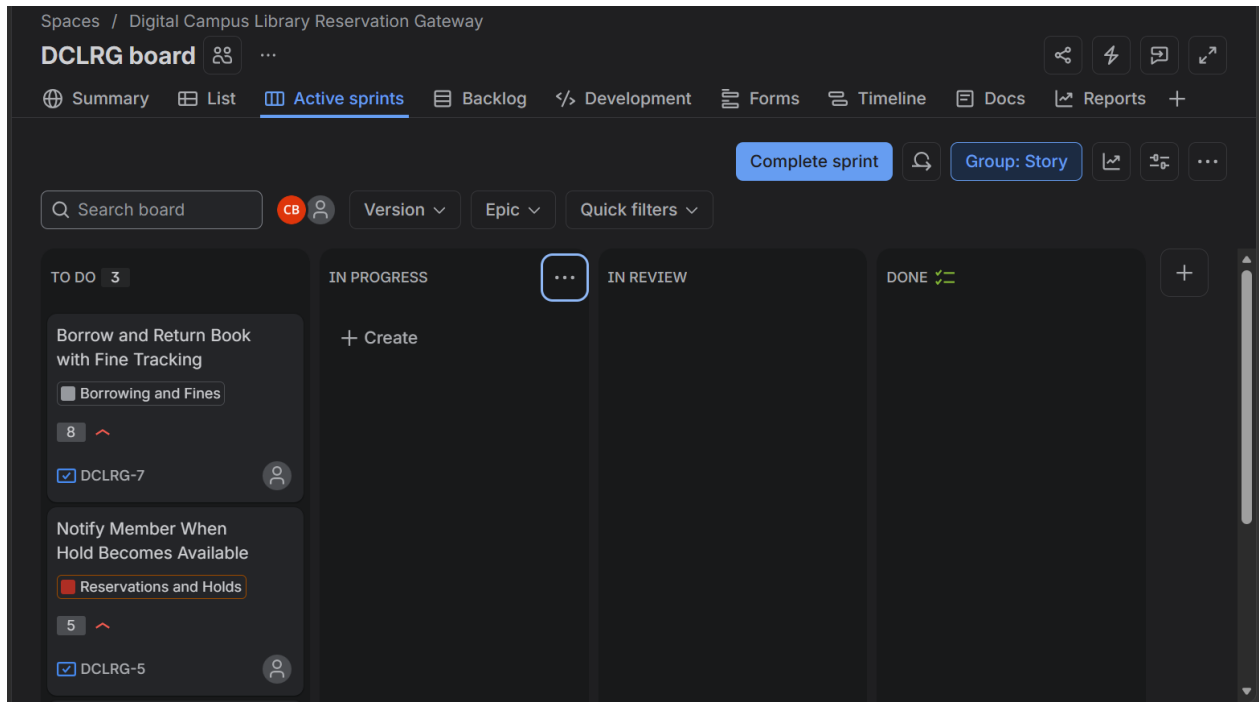
**Description**

As a Head Librarian, I want to add, edit, and retire ISBN catalog records and maintain the number of physical copies, so that the library catalog remains accurate and up to date.

**Attachments**

### 3. Sprint Execution - Sprint 1

Demonstrating the workflow progression across To Do, In Progress, In Review and Done from the same sprint1



Spaces / Digital Campus Library Reservation Gateway

**DCLRG board** ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Complete sprint Group: Story

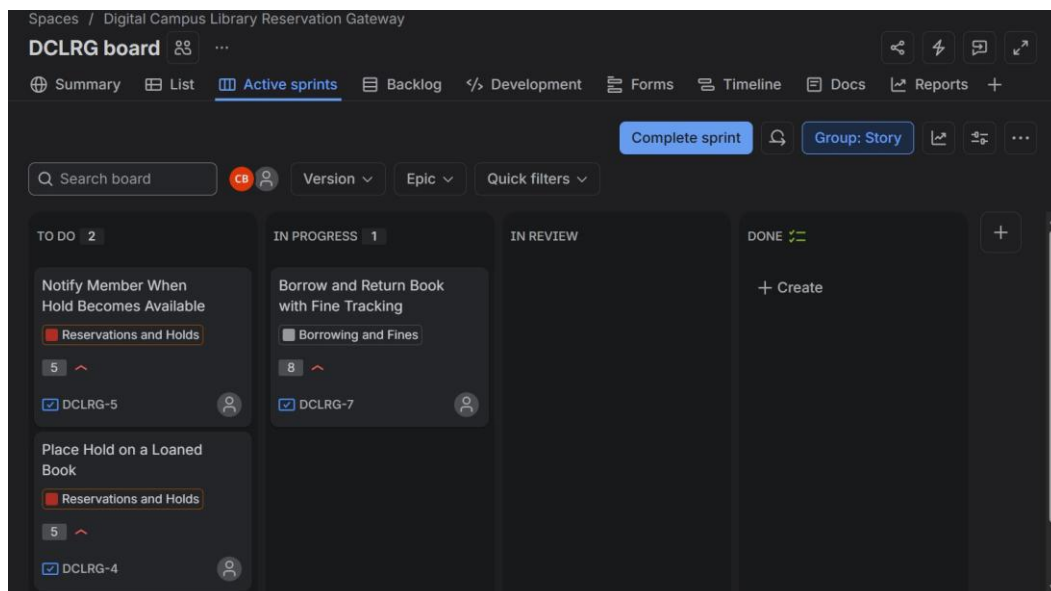
Search board CB Version Epic Quick filters

**TO DO 3**

- Borrow and Return Book with Fine Tracking
  - Borrowing and Fines
  - 8
  - ✓ DCLRG-7
- Notify Member When Hold Becomes Available
  - Reservations and Holds
  - 5
  - ✓ DCLRG-5

**IN PROGRESS** ... **IN REVIEW** **DONE** +

+ Create



Spaces / Digital Campus Library Reservation Gateway

**DCLRG board** ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Complete sprint Group: Story

Search board CB Version Epic Quick filters

**TO DO 2**

- Notify Member When Hold Becomes Available
  - Reservations and Holds
  - 5
  - ✓ DCLRG-5
- Place Hold on a Loaned Book
  - Reservations and Holds
  - 5
  - ✓ DCLRG-4

**IN PROGRESS 1** ... **IN REVIEW** **DONE** +

- Borrow and Return Book with Fine Tracking
  - Borrowing and Fines
  - 8
  - ✓ DCLRG-7

+ Create

Spaces / Digital Campus Library Reservation Gateway

**DCLRG board** ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Complete sprint Group: Story

Search board CB Version Epic Quick filters

**TO DO 1**

Place Hold on a Loaned Book

Reservations and Holds

5

DCLRG-4

+ Create

**IN PROGRESS 1**

Notify Member When Hold Becomes Avail...

Reservations and Holds

5

DCLRG-5

+ Create

**IN REVIEW**

**DONE 1**

Borrow and Return Book with Fine Tracking

Borrowing and Fines

8

DCLRG-7

Spaces / Digital Campus Library Reservation Gateway

**DCLRG board** ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Complete sprint Group: Story

Search board CB Version Epic Quick filters

**TO DO 1**

Place Hold on a Loaned Book

Reservations and Holds

5

DCLRG-4

+ Create

**IN PROGRESS**

**IN REVIEW 1**

Notify Member When Hold Becomes Available

Reservations and Holds

5

DCLRG-5

**DONE 1**

Borrow and Return Book with Fine Tracking

Borrowing and Fines

8

DCLRG-7

DCLRG board

Summary
List
Active sprints
Backlog
Development
Forms
Timeline
Docs
Reports

Complete sprint
Group: Story

Search board
CB
Version
Epic
Quick filters
Sprint details

TO DO 1

Place Hold on a Loaned Book

Reservations and Holds

5

☒ DCLRG-4

+ Create

IN PROGRESS

IN REVIEW

DONE 2

Notify Member When Hold Becomes Avail...

Reservations and Holds

5

☒ DCLRG-5

Borrow and Return Book with Fine Tracking

Borrowing and Fines

8

☒ DCLRG-7

Spaces / Digital Campus Library Reservation Gateway

DCLRG board

Summary
List
Active sprints
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Complete sprint
Group: Story

Search board
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Version
Epic
Quick filters

TO DO

+ Create

IN PROGRESS

IN REVIEW

DONE 3

Place Hold on a Loaned Book

Reservations and Holds

5

☒ DCLRG-4

Notify Member When Hold Becomes Available

Reservations and Holds

5

☒ DCLRG-5

## 4. Final Sprint State

After sprint-1

Spaces / Digital Campus Library Reservation Gateway

**DCLRG board** ...

Summary List Active sprints **Backlog** Development Forms Timeline Docs Reports +

Search backlog CB Version Epic Quick filters

**Epic** ×

- No epic
- > Catalog Management ...
- > Reservations and Holds ...
- > Borrowing and Fines ...
- + Create epic

**DCLRG Sprint 2** 1 Sep – 8 Sep (0 work items) 0 0 0 Start sprint ...

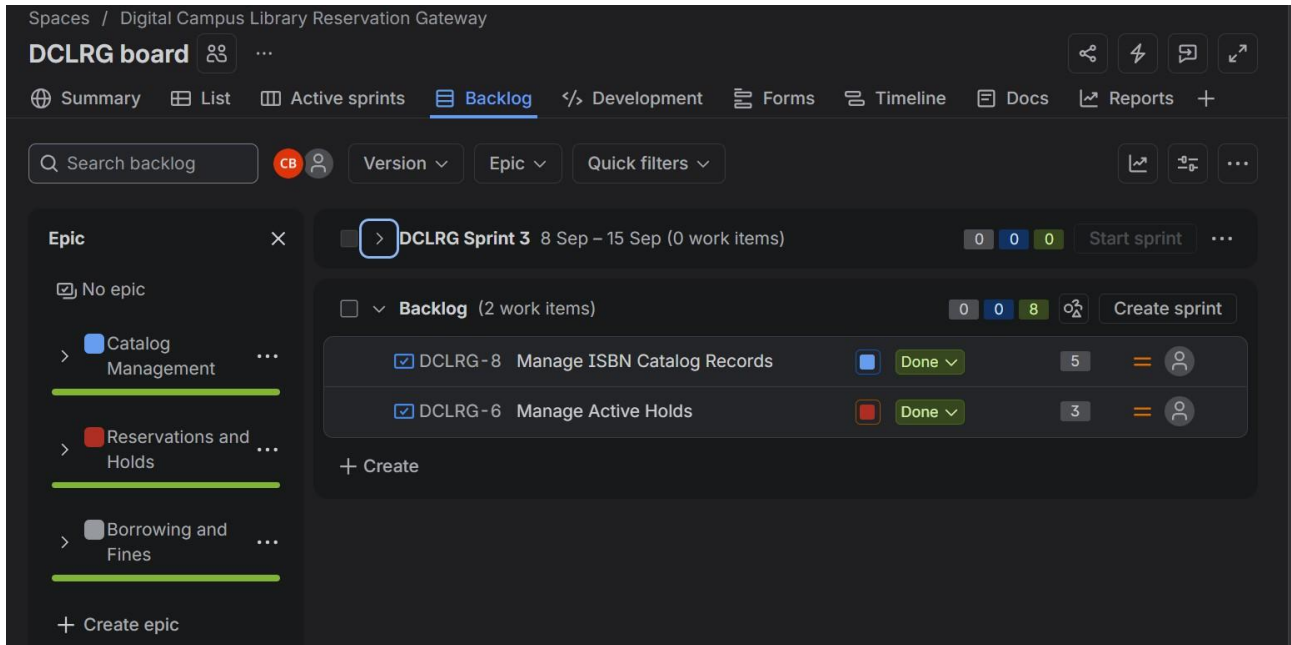
**Backlog** (5 work items) 8 0 18 Create sprint

Create a whiteboard to plan your work Try ... ×

|                                     |                                              |       |   |   |  |
|-------------------------------------|----------------------------------------------|-------|---|---|--|
| <input checked="" type="checkbox"/> | DCLRG-4 Place Hold on a Loaned Book          | Done  | 5 | ^ |  |
| <input checked="" type="checkbox"/> | DCLRG-5 Notify Member When Hold Become...    | Done  | 5 | ^ |  |
| <input checked="" type="checkbox"/> | DCLRG-7 Borrow and Return Book with Fine ... | Done  | 8 | ^ |  |
| <input checked="" type="checkbox"/> | DCLRG-6 Manage Active Holds                  | To Do | 3 | = |  |
| <input checked="" type="checkbox"/> | DCLRG-8 Manage ISBN Catalog Records          | To Do | 5 | = |  |

+ Create

## After sprint-2



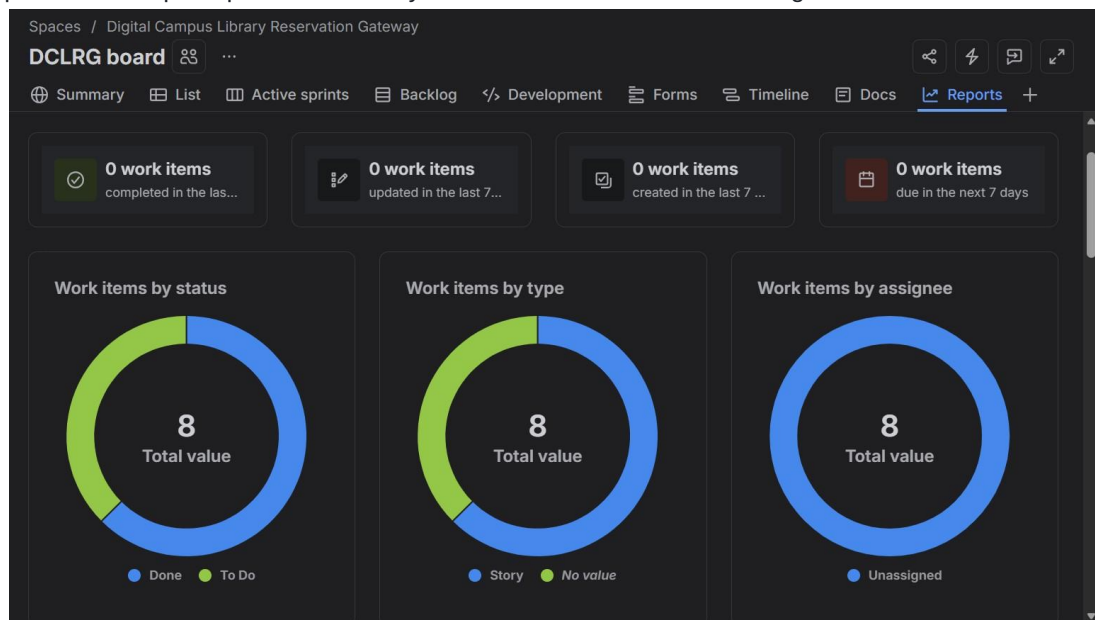
The screenshot shows the Jira DCLRG board in the Backlog view. The board is titled "DCLRG board" and is part of the "Digital Campus Library Reservation Gateway" space. The "Backlog" tab is selected, showing a list of work items. The "DCLRG Sprint 3" (8 Sep – 15 Sep) is currently active, with 0 work items. The backlog contains 2 work items:

| Work Item                           | Status | Priority | Assignee   |
|-------------------------------------|--------|----------|------------|
| DCLRG-8 Manage ISBN Catalog Records | Done   | 5        | [Assignee] |
| DCLRG-6 Manage Active Holds         | Done   | 3        | [Assignee] |

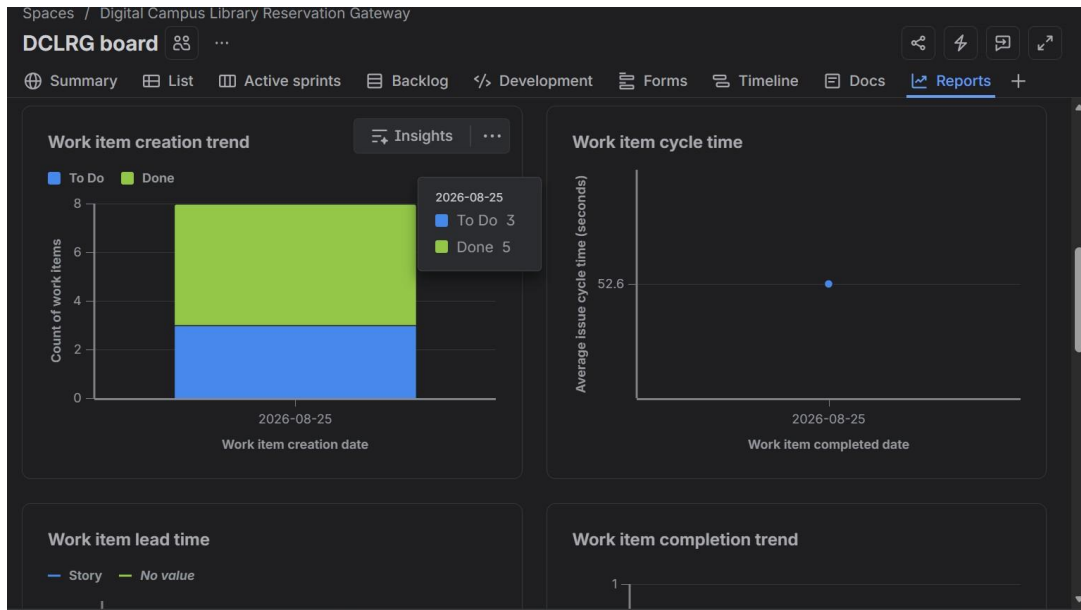
The left sidebar shows the "Epic" list with "Catalog Management", "Reservations and Holds", and "Borrowing and Fines". The bottom of the board shows a "Create" button.

## 5. Jira Analytics

The captured Jira Reports provide summary distributions and work-item timing information.

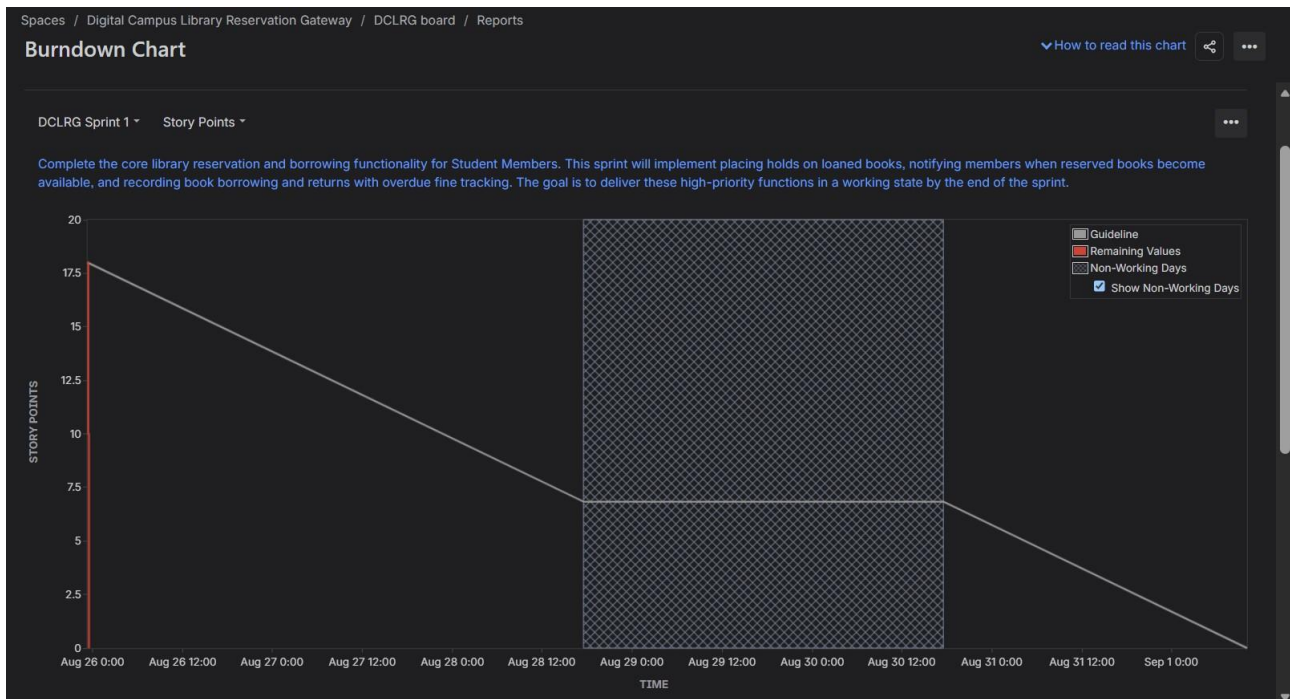






## 6. Sprint Analytics - Burndown Chart

The Burndown Chart is configured to use Story Points, matching the estimation method used for the Jira stories. The chart shows the remaining story-point workload against the guideline for DCLRG Sprint 1.



**Interpretation:** The Y-axis represents Story Points rather than Original Time Estimate, so the story-point values assigned to the user stories are reflected in the burndown.

## **7. Reflection**

### **1. Did the estimations reflect the actual effort?**

The Fibonacci story-point estimates provided a practical relative view of effort and complexity. The stories were successfully progressed through the planned workflow, although the captured analytics do not support a precise numerical comparison between points and actual effort.

### **2. Was the backlog well-prioritized?**

Yes. The backlog used explicit Jira priorities and story points. Higher-priority reservation, notification and borrowing functionality was selected for Sprint 1, while the remaining medium-priority management stories were completed in Sprint 2.

### **3. How did the simulated sprint align with the plan?**

The simulation followed the planned Scrum workflow. Sprint 1 contained three stories totaling 18 points, and Sprint 2 contained two stories totaling 8 points. Stories were moved through To Do, In Progress, In Review and Done.

### **4. What insights did the burndown chart give about team capacity?**

The burndown chart provides a visual comparison between the remaining story points and the ideal guideline. Because this was a simulated lab sprint, the chart is mainly useful for demonstrating how planned scope and completion progress are monitored.